

Additional Information

Scientific Contributions

JOURNAL ARTICLES

- **Gomariz A**, Kikuchi Y, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Valcarcel AM, Luu J, Ferrara D, Goksel O, “Joint semi-supervised and contrastive learning enables zero-shot and multi-domain segmentation”, *Medical Image Analysis* (2024, in review)
- Yamada Y, Nguyen TT, Impellizzieri D, Mineura D, Shibuya R, **Gomariz A**, Haberecker M, Nilsson J, Nombela-Arrieta C, Junggraithmayr W, Boyman O “Biased IL-2 signals induce Foxp3-rich pulmonary lymphoid structures and facilitate long-term lung allograft acceptance in mice”, *Nature Communications* 1383 (14) (2023). doi: [10.1038/s41467-023-36924-z](https://doi.org/10.1038/s41467-023-36924-z)
- Helbling PM, Vijaykumar A, **Gomariz A**, Zerkatje T, Zielińska KA, Loosli K, Isringhausen S, Nagasawa T, Roeder I, Manz MG, Yokomizo T, Nombela-Arrieta C “Tissue-scale dynamic mapping of Hematopoietic Stem Cells and supportive niche cells in the fetal liver”, *bioRxiv* (2023). doi: [10.1101/2023.09.12.554625](https://doi.org/10.1101/2023.09.12.554625)
- **Gomariz A**, Portenier T, Nombela-Arrieta C, Goksel O, “Probabilistic Spatial Analysis in Quantitative Microscopy with Uncertainty-Aware Cell Detection using Deep Bayesian Regression”, *Science Advances* 8 (5) (2022). doi: [10.1126/sciadv.abi8295](https://doi.org/10.1126/sciadv.abi8295)
- **Gomariz A**, Portenier T, Helbling PM, Isringhausen S, Suessbier U, Nombela-Arrieta C, Goksel O, “Modality Attention and Sampling Enables Deep Learning with Heterogeneous Marker Combinations in Fluorescence Microscopy”, *Nature Machine Intelligence* p. 1-13 (2021). doi: [10.1038/s42256-021-00379-y](https://doi.org/10.1038/s42256-021-00379-y)
- Isringhausen S, Mun Y, Kovtonyuk L, Krautler N, Suessbier U, **Gomariz A**, Spaltro G, Helbling PM, Wong HC, Nagasawa T, Manz MG, Oxenius A, Nombela-Arrieta C, “Chronic viral infections persistently alter marrow stroma and impair hematopoietic stem cell fitness”, *Journal of Experimental Medicine* 218 (12) (2021). doi: [10.1084/jem.20192070](https://doi.org/10.1084/jem.20192070)
- Ghazi S, Bourgeois S, **Gomariz A**, Bugarski M, Haenni D, Martins JR, Nombela-Arrieta C, Wagner CA, Hall AM, Eilidh C, “Multiparametric imaging reveals that mitochondria rich intercalated cells in the kidney collecting duct have a very high glycolytic capacity”, *The FASEB journal* 34 (6), pp. 8510–8525 (2020). doi: [10.1096/fj.202000273R](https://doi.org/10.1096/fj.202000273R)
- **Gomariz A**, Isringhausen S, Helbling PM, and Nombela-Arrieta C, “Imaging and spatial analysis of hematopoietic stem cell niches”, *Annals of the New York Academy of Sciences* 1466(1):5-16 (2019). doi: [10.1111/nyas.14184](https://doi.org/10.1111/nyas.14184)
- Hess M, **Gomariz A**, Goksel O, Ewald CY, “In-vivo quantitative image analysis of age-related morphological changes of *C. elegans* neurons reveals a correlation between neurite bending and novel neurite outgrowths”, *eNeuro* 6(4) (2019). doi: [10.1523/ENEURO.0014-19.2019](https://doi.org/10.1523/ENEURO.0014-19.2019)
- Agarwal P, Isringhausen S, Li H, Paterson AJ, He J, **Gomariz A**, Nagasawa T, Nombela-Arrieta C, and Bhatia R, “Mesenchymal niche-specific expression of Cxcl12 controls quiescence of treatment-resistant leukemia stem cells”, *Cell Stem Cell* 24 (5), 769–784.e6 (2019). doi: [10.1016/j.stem.2019.02.018](https://doi.org/10.1016/j.stem.2019.02.018)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Quantitative spatial analysis of haematopoiesis- regulating stromal cells in the bone marrow microenvironment by 3D microscopy”, *Nature Communications* 9 (1) (2018). doi: [10.1038/s41467-018-04770-z](https://doi.org/10.1038/s41467-018-04770-z)
- Takizawa H, Fritsch K, Kovtonyuk LV, Saito Y, Yakkala C, Jacobs K, Ahuja AK, Lopes M, Hausmann A, Hardt WD, **Gomariz A**, Nombela-Arrieta C, and Manz MG, “Pathogen-Induced TLR4-TRIF Innate Immune Signaling in Hematopoietic Stem Cells Promotes Proliferation but Reduces Competitive Fitness”, *Cell Stem Cell* 21 (2), 225–240.e5 (2017). doi: [10.1016/j.stem.2017.06.013](https://doi.org/10.1016/j.stem.2017.06.013)

CONFERENCE PAPERS

- Bredell G, Fischer M, Szostak P, Abbasi-Sureshjani S, and **Gomariz A**, “The Importance of Downstream Networks in Dig-

ital Pathology Foundation Models”, *MICCAI - MedAGI Workshop* (2024, accepted)

- Ibañez v, Szostak P, Wong Q, Korski K, Abbasi-Sureshjani S, and **Gomariz A**, “Integrating multiscale topology in digital pathology with pyramidal graph convolutional networks”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1-4 (2024). doi: [10.1109/ISBI56570.2024.10635702](https://doi.org/10.1109/ISBI56570.2024.10635702)
- Kikuchi Y, **Gomariz A**, Li Y, Lu H, Albrecht A, Ferrara D, “Device adaptation of optical coherence tomography (OCT) retinal layer segmentation algorithm using unlabeled target data”, *Investigative Ophthalmology & Visual Science, ARVO abstract* (2023)
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Valcarcel AM, Luu J, Ferrara D, Goksel O, “Unsupervised Domain Adaptation with Contrastive Learning for OCT Segmentation”, *International Conference on Medical image computing and computer-assisted intervention (MICCAI)* pp. 351-361 (2022). doi: [10.1007/978-3-031-16452-1_34](https://doi.org/10.1007/978-3-031-16452-1_34)
- Li Y, Lu H, Albrecht T, Maunz A, Neubert A, Benmansour F, **Gomariz A**, Luu J, Ferrara D “Deep learning (DL) to segment retinal layer disruption on optical coherence tomography (OCT) in neovascular age-related macular degeneration (nAMD)”, *Investigative Ophthalmology & Visual Science, ARVO abstract* (2022)
- **Gomariz A**, Lu H, Li Y, Albrecht A, Maunz A, Benmansour F, Luu J, Goksel O, Ferrara D, “A unified deep learning approach for OCT segmentation from different devices and retinal diseases”, *Investigative Ophthalmology & Visual Science, ARVO abstract* (2022)
- **Gomariz A**, Egli R, Portenier T, Nombela-Arrieta C, and Goksel O, “Utilizing Uncertainty Estimation in Deep Learning Segmentation of Fluorescence Microscopy Images with Missing Markers”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 371-374 (2021). doi: [10.1109/ISBI48211.2021.9434158](https://doi.org/10.1109/ISBI48211.2021.9434158)
- **Gomariz A**, Li W, Ozkan E, Tanner C, and Goksel O, “Siamese Networks with Location Prior for Landmark Tracking in Liver Ultrasound Sequences”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 1757-1760 (2019). doi: [10.1109/ISBI.2019.8759382](https://doi.org/10.1109/ISBI.2019.8759382)
- Goksel O, Vishnevsky V, **Gomariz A**, and Tanner C, “Imaging of Sliding Visceral Interfaces During Breathing”, *IEEE International Symposium on Biomedical Imaging (ISBI)* pp. 298-301 (2016). doi: [10.1109/ISBI.2016.7493268](https://doi.org/10.1109/ISBI.2016.7493268)

ORAL PRESENTATIONS AT CONFERENCES

- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Goksel O, and Nombela-Arrieta C, “A Convolutional Neural Network-Based Pipeline for the Characterization of Large Bone Marrow Images in 3D Microscopy”, *Swiss Light-Sheet Microscopy Workshop, Zurich* (2019)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Goksel O, Manz MG, and Nombela-Arrieta C, “Quantitative Characterization of Bone Marrow Stroma Using Deep Learning”, *Day of Clinical Research, University Hospital of Zurich* (2018)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Quantitative characterization of bone marrow stromal topography using deep convolutional neural networks”, *Seeing is Believing – Imaging the Molecular Processes of Life, EMBL, Heidelberg* (2017)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Paul G, Székely G, Stoma S, Nørrelykke SF, and Nombela-Arrieta C, “Applying 3D quantitative microscopy to study global topography and cellular interactions in the bone marrow”, *Image-based Systems Biology, Jena* (2016)

INVITED TALKS

- “Quantitative Fluorescence Microscopy with Marker-Efficient and Uncertainty-Aware Segmentation and Detection Using Deep Learning”, *Visual Information and Interaction Seminar, Uppsala University* (2021)
- “Microscopy-based characterization of the bone marrow structure using deep learning and spatial statistics”, *Joint Immunology Meeting, University of Zurich and ETH Zurich* (2020)
- “Image-based characterization of deep tissue fluorescence microscopy using convolutional neural networks”, *Institute of Molecular Health Sciences, ETH Zurich* (2019)

- “Deep learning methods for image-based understanding of the bone marrow microenvironment”, *Roche, Basel* (2019)
- “Image-based characterization of the bone marrow stroma using convolutional neural networks and spatial statistics”, *Center for Microscopy and Image Analysis, University of Zurich* (2018)
- “Quantitative spatial analysis of bone marrow tissues using 3D microscopy and deep learning”, *Joint Cancer Meeting, University of Zurich and ETH Zurich* (2018)

POSTER PRESENTATIONS AT CONFERENCES

- **Gomariz A**, Lu H, Li Y, Albrecht T, and Goksel O, “Deep learning with labeled and unlabeled optical coherence tomography images”, *Academia meets Industry: Basel Biozentrum / Roche Innovation Symposium* (2021)
- **Gomariz A**, Lu H, Li Y, Albrecht T, and Goksel O, “Deep semi-supervised learning for image segmentation in multi-domain optical coherence tomography”, *Basel Postdoc Network Meeting, Kandersteg* (2021)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Goksel O, Manz MG, and Nombela-Arrieta C, “A deep learning pipeline for spatial characterization of bone marrow stroma”, *14th Charles Rodolphe Brupbacher Symposium, Zurich* (2019)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Spatial Analysis of the Bone Marrow Stroma Using Deep Learning”, *Swiss Oncology & Hematology Congress, Zurich* (2018)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Convolutional Neural Networks in 3D Microscopy: Spatial Analysis of the Bone Marrow Microenvironment”, *2nd NEUBIAS Conference, Szeged* (2018)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Multiscale image-based quantitative analysis of bone marrow stromal network topology reveals strict spatial constraints for hematopoietic-stromal cellular interactions”, *4th Lymphoid Tissue Meeting, St. Gallen* (2017)
- **Gomariz A**, Helbling PM, Isringhausen S, Suessbier U, Becker A, Boss A, Nagasawa T, Paul G, Goksel O, Székely G, Stoma S, Nørrelykke SF, Manz MG, and Nombela-Arrieta C, “Applying 3D quantitative microscopy to study global topography and cellular interactions in the bone marrow”, *Charles Rodolphe Brupbacher Symposium, Zurich* (2017)
- **Gomariz A**, Remacha E, Vizcaino G, Corral A, Paarman M, Zacharakis G, Vaquero JJ, Desco M, Moscoso M, and Ripoll J, “Protocol for Establishing the Resolution in Laser Sheet Microscopes”, *European Molecular Imaging Meeting, Leuven* (2014)
- **Gomariz A**, Remacha E, Vizcaino G, Corral A, Paarman M, Vaquero JJ, Desco M, Moscoso M, Scheffold F, and Ripoll J, “Resolution Comparison between Laser Sheet Microscopy and Optical Projection Tomography”, *European Molecular Imaging Meeting, Leuven* (2014)

Awards and Scholarships

EXCITE PhD Excellence Award for outstanding PhD thesis in the field of biomedical imaging	2021
Hasler Stiftung grant to fund PhD project for 1 year	2019
Award to the best presented project at the 17 th Day of Clinical Research at University Hospital of Zurich, in the category Regenerative Medicine and Advanced Technologies	2018
Willi Studer Prize conferred to the student who attained the highest overall grade in the MSc in Biomedical Engineering at ETH Zurich	2017
Postgraduate Scholarship granted by Fundación Caja Madrid to undertake a Master's degree at ETH Zurich	2014
Extraordinary End of Studies Award conferred to the student who attained the highest overall grade in the BSc in Biomedical Engineering at University Carlos III of Madrid	2014
Bachelor thesis awarded with special distinction for the best thesis of the graduating class	2014
Finalist in the StartUp Programme competition: SAGE Award to the entrepreneur project eMaging	2014
Excellence Award 2014 granted to the 20 students with best grades that year in the University Carlos III of Madrid	2014
Travel grant for presenting a poster and assisting to EMIM 2014 in Belgium	2014
Collaboration scholarship for conducting research at the University Carlos III of Madrid	2013
Scholarship for new students with best entry grades at University Carlos III of Madrid	2010
Scholarship for graduating high school with distinction (top 5%) at IES Ramón Arcas Meca, Spain	2010